LeMETER:

Integrating LeJEPA Representation Learning with MobileViT-Based METER for Monocular Depth Estimation

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Problem Statement

Monocular depth estimation in robotics requires :

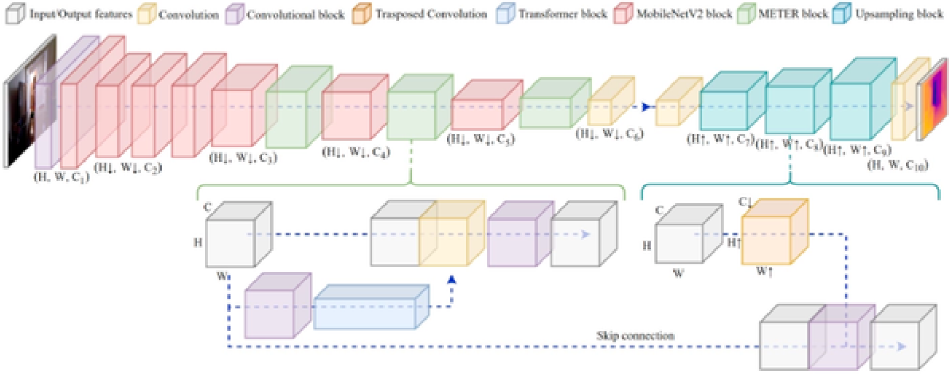
- Lightweight
- Fast
- Accuracy

The challenge is to improve depth prediction quality while keeping the model compact and efficient.

State of the Art

METER - SOTA lightweight model for monocular depth estimation

- MobileViT-based
- Balance accuracy, inference speed, and model size - 0.71M-3.29M.

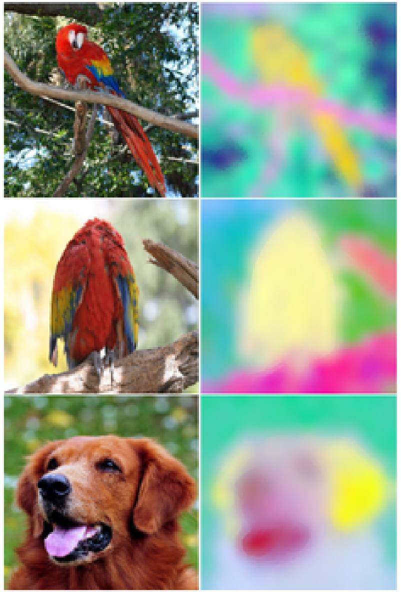


Models	NYU RMSE↓ [m]	NYU REL↓	NYU δ_1 ↑	NYU MAC [G]
CReaM [32]	0.687	0.190	0.704	-
PyD-Net (50) [31]	-	-	-	-
PyD-Net (200) [31]	-	-	-	-
FastDepth [8]	0.579	-	0.772	3.210
M.Net v2 + FBNet [34]	0.564	-	0.790	-
SPEED [7]	0.566	0.158	0.783	0.552
METER S	**0.471**	**0.134**	**0.831**	0.975
METER XS	0.522	0.154	0.793	0.579
METER XXS	0.580	0.174	0.744	0.186

State of the Art

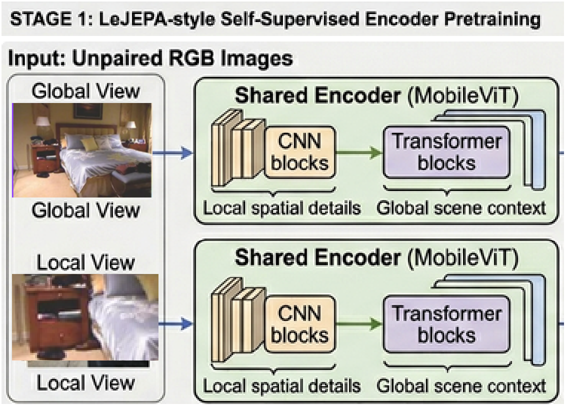
Lejepa - a methods for strong self-supervised visual representations

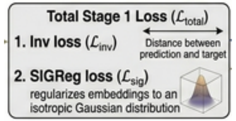
- The isotropic Gaussian - model work well on every downstream task
- SIGReg — enforce isotropic Gaussian distribution in high-dimensional space in a stable and scalable way

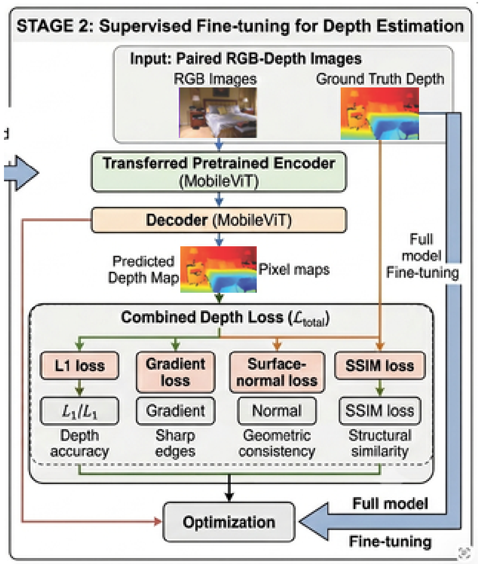


Proposed Method

LeMETER, combining **LeJEPA**-inspired representation learning with the **METER** architecture



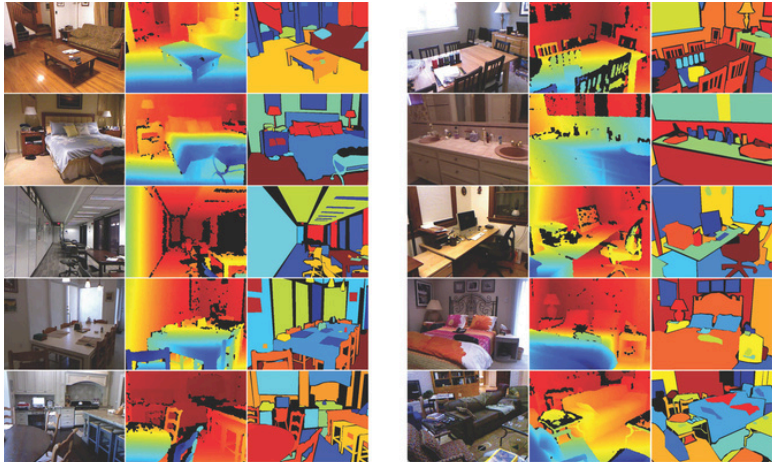




Dataset

NYU Depth V2

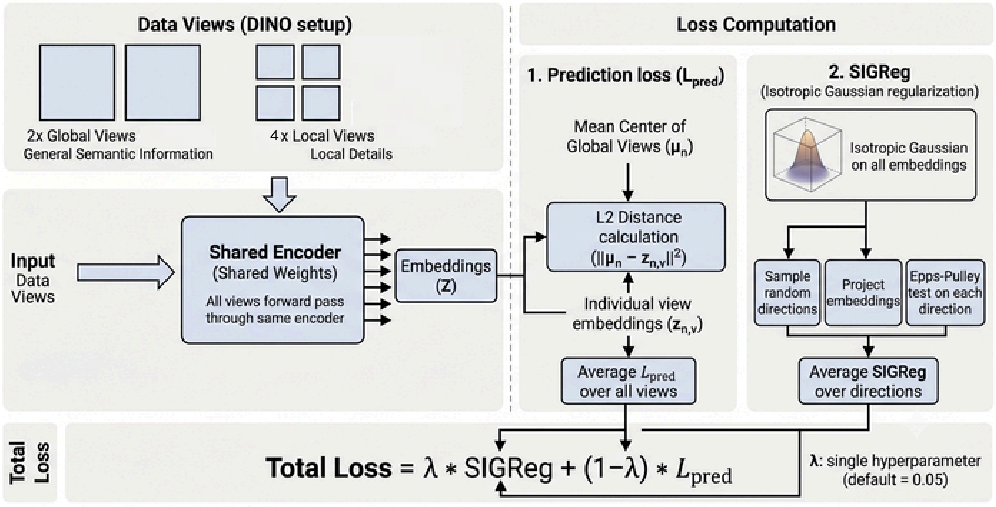
- An indoor RGB-D dataset containing aligned RGB images and depth maps.
- The depth maps have a maximum distance of 10 meters.
- For this experiment, I used the 50K subset.
- Input size as 256x192



Experimental Setup

Pretrain METER encoder with LeJEPA method

- Using 6 views for training
  - 2 global and 4 local views
- Projection dim: 64
- Hidden dim: 256
- Batch size: 128
- Lr: 1e-3
- lambda sigreg: 0.05
- local crop: 0.25-0.6
- global crop: 0.5-1
- epoch: 20
- Loss = lambda*sigreg + (1-lambda) * inv



Experimental Setup

METER

Hyperparameters are based on the original paper except:

- Loss in origin repo: **Ldepth / lambda1** instead of **Ldepth *lambda1** like in the paper
- Remove random crop
  - → reduce training time from 12m to 4m per epoch
- After epoch 8, add extra augumentation

Hardware: Kaggle Tesla T4

HPara	METER	LeMETER
LR schedule	Decay learning rate by 0.1 every 20 epochs.	cosine annealing 2e-3 - 1e-4
Augmentation	Vertical flip, mirroring, random crop, channel swap, Brightness/gamma, color scaling factor, depth shift.	No random crop, vertical flip. Extra: Gaussian blur, gray scale, color brightness, contrast, saturation
Regularization		Drop out, path drop out, attention drop out.

Model Evaluation

Metrics

- RMSE (Root Mean Squared Error)
  - Average squared difference
  - Lower is better.
  - Sensitive to large errors.
- AbsRel (Absolute Relative Error)
  - Measures relative depth error.
  - Lower is better.
  - Less sensitive to faraway objects than RMSE.
- δ1 (Threshold Accuracy)
  - Percentage of pixels where the ratio is within 1.25.
  - Higher is better.

$$RMSE = \sqrt{\frac{1}{|n|}\sum_{i \in n} ||y_i - \hat{y}_i||^2}$$

$$REL = \frac{1}{|n|}\sum_{i \in n} \frac{|y_i - \hat{y}_i|}{y_i}$$

$$\delta_1 = \frac{1}{|n|}\sum_{i \in n} \max\left(\frac{y_i}{\hat{y}_i}, \frac{\hat{y}_i}{y_i}\right) < thr$$

Model Evaluation

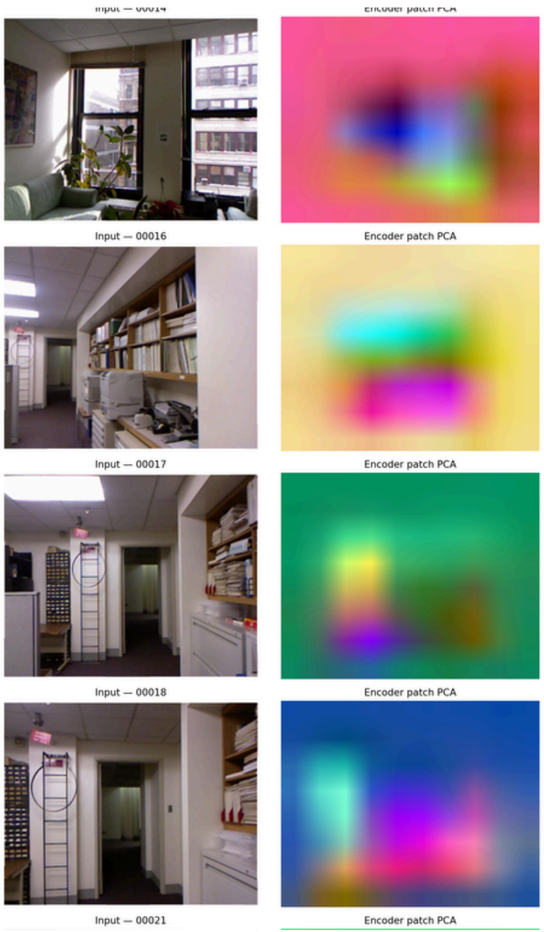
LeJEPA pretraining stage

Zero-shot Geometric Probing

- Center-biased
- Some coarse foreground/background separation in a few samples,
- Contours do not line up tightly with depth discontinuities

Reasons:

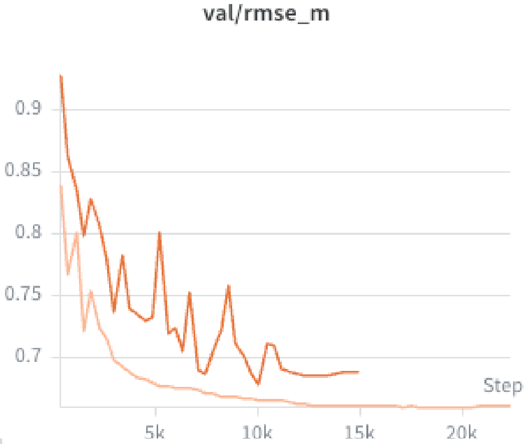
- Lejepa treats global views and local views as image-level representations
- While depth senses require dense spatial features

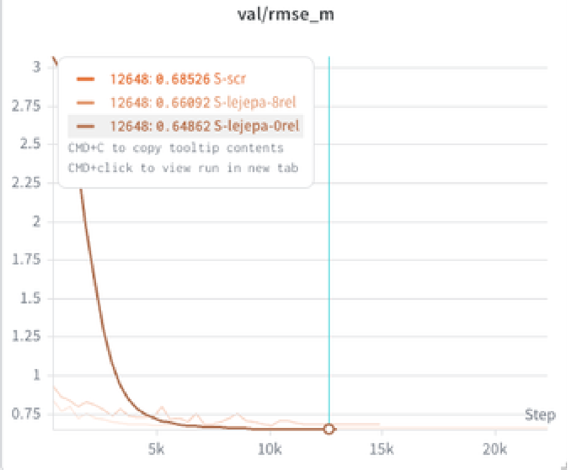


Model Evaluation

Finetuning Stage

- METER finetuned with lejepa is smother, faster and better accuracy
- The LeMeter combined with regularization started with a very high loss, but this loss decreased quickly, allowing it to outperform METER without regularization.





METER from scratch

LeMETER

LeMETER with dropout

Model Evaluation

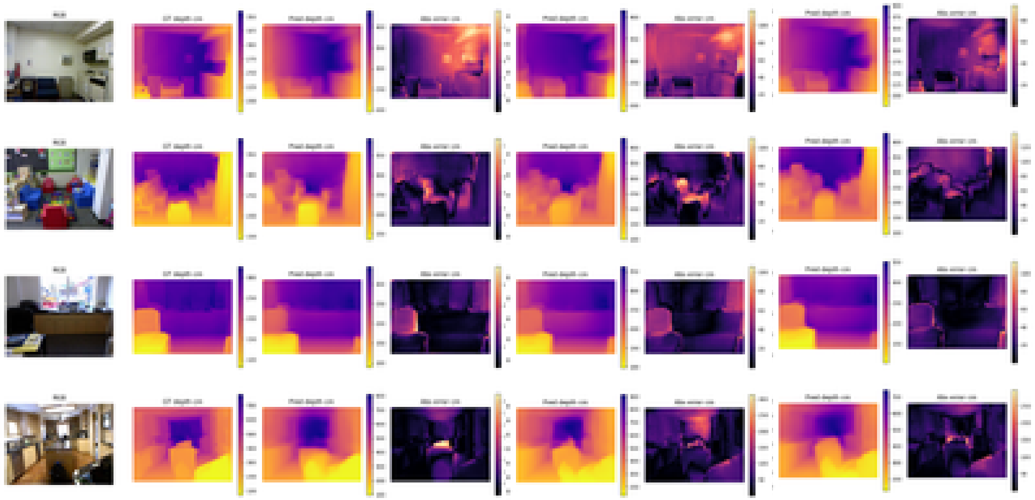
Result

- Rel and del between the models are not much different.
- But in RMSE, there is a noticeable gap between the models
- LeMETER out performce METER from scratch model but it took extra 5h for pretraining encoder

Model	RMSE ↓	REL ↓	δ1 ↑
METER scratch	0.685	**0.163**	0.765
LEMETER	0.659	**0.163**	0.773
LEMETER + dropout	**0.647**	0.164	**0.776**

Model Evaluation

Visualization



Ground True

METER scratch

LEMETER

LEMETER + dropout

Conclusions

- The pretrained encoder
  - PCA show not much depth continuities alignment
  - But it still speed up the training process and produced a better result
- Training cost needs to be considered. Lejepa pretraining will be suitable for:
  - Limited data
  - Long training session

Future work

- Explore other Jepa variants like V-Jepa2.1 or combine them with Lejepa for better local feature extraction
- Additional regularization and augmentation methods

References

- Papers and images:
  - Balestriero, R., & LeCun, Y. (2025). LeJEPA: Provable and Scalable Self-Supervised Learning Without the Heuristics.
  - Papa, L., Russo, P., & Amerini, I. (2024). METER: A Mobile Vision Transformer Architecture for Monocular Depth Estimation.

- Slides
  - https://github.com/pietro-nardelli/sapienza-ppt-template
  - [Attribution-NonCommercial-ShareAlike4.0International(CC BY-NC-SA 4.0)]

Thank you for the attention!